Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/02/2023
PI Name:	Koehne, Jessica Ph.D.		
Project Title:	Printed Electrochemical Sensor Strip for Quantifying Bone Density Loss in Microgravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Jessica.e.koehne@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-6818
Organization Name:	NASA Ames Research Center		
PI Address 1:	Center for Nanotechnology		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	10/01/2020	**End Date:**	10/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/30/2022 per PI (Ed., 8/25/22)		
Key Personnel Changes/Previous PI:	July 2021 report: Milton Cordeiro is now Co-Investigator on the project.		
COI Name (Institution):	Graf, John Ph.D. (NASA Johnson Space Center) Cordeiro, Milton Ph.D. (USRA)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Human Research Program has outlined risks associated with human spaceflight within the Human Research Roadmap. One such risk is listed as the risk of bone fracture due to spaceflight-induced changes to bone. Our objective to create a printed amino-terminal collagen crosslinks (NTX) quantification sensor strip coupled with a facile urine collection and volume measurement device. The proposed project will be a ground-based study with the potential to be further developed for spaceflight. To accomplish the proposed objective, we will complete the following Aims. Aim 1: Develop conductive, telopeptide selective, and dielectric inks for printed sensor. Aim 2: Print and electrochemically characterize 3-electrode device for NTX detection. Aim 3: Integrate sensor with urine collection device and handheld potentiostat hardware. If successful, the proposed project will reduce risk of crew bone fracture by continuously evaluating bone health by monitoring mineral metabolism as excreted NTX for bone reabsorption. Future studies could expand the scope of health monitoring to include interferon gamma, tumor necrosis factor-alpha, 25 OH-vitamin B, and bone specific alkaline phosphatase, and/or other molecules of interest. Additionally, these sensors will be manufactured entirely by printing technology. It is anticipated that they can eventually be manufactured in an in-space environment, which directly compliments Space Technology Mission Directorate's In-Space Manufacturing project. By relying on simple printing technology, analytical sensors can be fabricated in space, which would enable adaptive crew health monitoring on long-duration space mission and future habitation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has benefits for both humans involved in space travel and for humans on Earth. A sensor that monitors bone health has great benefit for our aging adult population. Osteoporosis is the most common bone disease and affects an estimated 10.2 million Americans leading to increased risk of bone fracture. Point-of-care sensors can be useful to monitor bone density loss and onset of osteoporosis, improving the patient experience by lowering the time required for an auxiliary diagnostic while also reducing healthcare costs, since a specialized operator is not required to perform the measurement.
Task Progress:	This project addresses the topic: "Lab Analysis Point-of-Care Device Evaluation and Downselect" of the Human Research Roadmap (HRR) gap Osteo 5 that states, "we need an in-flight capability to monitor bone turnover and bone mass changes during spaceflight." One bone remodeling biomarker that can be directly correlated with bone resorption from a non-invasive urine specimen is NTx, a degradation by-product of type I collagen. NTx levels in urine were recently used to evaluate bone health during the NASA Twin Study and is a biomarker of interest, referred to in HRR's Osteo 5 gap. To address this HRR gap, we have developed a methodology using a printed electrochemical sensor capable of detecting NTx in urine. Our goal is to create a highly adaptable and versatile approach that utilizes fabrication processes consistent with in-space manufacturing, thus enabling the manufacture of point-of-care devices during flight. Going forward, we plan to expand the sensor's capability to measure a variety of bone remodeling biomarkers simultaneously. However, NTx serves as our initial target for this project and as a proof-of-concept for our approach. The project objective is to create a space-suitable, printed sensor strip for quantifiable NTx detection coupled to a urine collection device. To accomplish this objective, we have focused on the following Aims. Aim 1: Develop conductive, telopeptide selective and dielectric inks for printed sensor. Aim 2: Print and electrochemically characterize 3-electrode device for NTx detection. Aim 3: Integrate sensor with urine collection device and handheld potentiostat hardware. During this project, we have created a hands-free and robust manufacturing process using microdispense, inkjet, and 3D printing to generate an electrochemical sensor for measuring NTx from a urine specimen. The sensor utilizes a carbon nanotube ink, which improves the electrochemical response and gold nanoparticles to provide selective surfaces for NTx detection. Using those optimized and pre-screened electronic inks, 3-electrode electrochemical sensors have been printed using inkjet and precision microdispense instrumentation. Both devices were characterized for their electrochemical performance and robustness. Fluid delivery based on lateral flow materials was designed to carry the urine specimen to the sensor surface. Finally, sensors were packaged into reusable cassette holders and interfaced with a handheld reader for simple and reliable operation.
Bibliography Type:	Description: (Last Updated: 04/14/2023)
Abstracts for Journals and Proceedings	Koehne, J. E., Cordeiro, M. "Printed electrochemical sensor for quantifying bone density loss in microgravity." 2021 NASA Human Research Program Invesitgators' Workshop, virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Koehne JE. "Printed flexible sensors for NASA applications." 4DMS + SoRo Summer School, virtual, June 21-22, 2021. Abstracts. 4DMS + SoRo Summer School, virtual, June 21-22, 2021. , Jun-2021
Abstracts for Journals and Proceedings	Koehne JE. "In-space manufacturing of point-of-care diagnostic devices." American Chemical Society CME NASA STEM Symposium, Atlanta, Georgia, August 23-24, 2021. Abstracts. American Chemical Society CME NASA STEM Symposium, Atlanta, Georgia, August 23-24, 2021. , Aug-2021
Abstracts for Journals and Proceedings	Koehne JE. "Printed sensors and electronics for sustained human spaceflight." First International Conference on Smart Green Connected Societies, virtual, November 29-30, 2021. Abstracts. First International Conference on Smart Green Connected Societies, virtual, November 29-30, 2021. , Nov-2021
Abstracts for Journals and Proceedings	Koehne JE. "In-space manufacturing to support human spaceflight. " Materials Research Society, Boston, Massachusetts, November 28-December 2, 2021. Abstracts. Materials Research Society, Boston, Massachusetts, November 28-December 2, 2021. , Nov-2021

Abstracts for Journals and Proceedings	Koehne JE. "Carbon nanomaterial based electrochemical sensors and devices for NASA missions." IEEE NMDC, Vancouver, Canada, December 12-15, 2021. Abstracts. IEEE NMDC, Vancouver, Canada, December 12-15, 2021. , Dec-2021
Abstracts for Journals and Proceedings	Cordeiro M, Espinoza C, Koehne JE. "Printed electrochemical sensor for quantifying bone density loss in microgravity." 2022 NASA Human Research Program Investigators' Workshop, virtual, February 8, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, virtual, February 8, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Cordeiro M. "Printed wearable electrochemical sensor for monitoring human performance markers during human spaceflight." Electrochemical Society Meeting, Vancouver, Canada, May 31, 2022. Abstracts. Electrochemical Society Meeting, Vancouver, Canada, May 31, 2022. , May-2022
Abstracts for Journals and Proceedings	Koehne JE. "Bioanalytical sensors for support human exploration missions." Gordon Research Conference on Bioanalytical Sensors, Newport, Rhode Island, June 26, 2022. Abstracts. Gordon Research Conference on Bioanalytical Sensors, Newport, Rhode Island, June 26, 2022. , Jun-2022
Articles in Other Journals or Periodicals	Cordeiro M, Koehne JE. "Fully inkjet printed carbon nanotube electrochemical sensor for NASA missions." ECS Sensors Plus, submitted. , Jan-2023